is used in *future* calculations. (The term "recursive" also has other meanings, especially in computer science). Chapter 19 discusses a variety of recursive filters in more detail. Be aware that the moving average recursive filter is very different from typical recursive filters. In particular, most recursive filters have an infinitely long impulse response (IIR), composed of sinusoids and exponentials. The impulse response of the moving average is a rectangular pulse (finite impulse response, or FIR).

This algorithm is faster than other digital filters for several reasons. First, there are only two computations per point, regardless of the length of the filter kernel. Second, addition and subtraction are the only math operations needed, while most digital filters require time-consuming multiplication. Third, the indexing scheme is very simple. Each index in Eq. 15-3 is found by adding or subtracting integer constants that can be calculated before the filtering starts (i.e., p and q). Forth, the entire algorithm can be carried out with integer representation. Depending on the hardware used, integers can be more than an order of magnitude faster than floating point.

Surprisingly, integer representation works *better* than floating point with this algorithm, in addition to being *faster*. The round-off error from floating point arithmetic can produce unexpected results if you are not careful. For example, imagine a 10,000 sample signal being filtered with this method. The last sample in the filtered signal contains the accumulated error of 10,000 additions and 10,000 subtractions. This appears in the output signal as a drifting offset. Integers don't have this problem because there is no round-off error in the arithmetic. If you must use floating point with this algorithm, the program in Table 15-2 shows how to use a double precision accumulator to eliminate this drift.

```
100 'MOVING AVERAGE FILTER IMPLEMENTED BY RECURSION
110 'This program filters 5000 samples with a 101 point moving
120 'average filter, resulting in 4900 samples of filtered data.
130 'A double precision accumulator is used to prevent round-off drift.
140 '
150 DIM X[4999]                    'X[ ] holds the input signal
160 DIM Y[4999]                    'Y[ ] holds the output signal
170 DEFDBL  ACC                    'Define the variable ACC to be double precision
180 '
190 GOSUB XXXX                     'Mythical subroutine to load X[ ]
200 '
210 ACC = 0                        'Find Y[50] by averaging points X[0] to X[100]
220 FOR I% = 0 TO 100
230   ACC = ACC + X[I%]
240 NEXT I%
250 Y[[50] = ACC/101
260 '                              'Recursive moving average filter (Eq. 15-3)
270 FOR I% = 51 TO 4949
280   ACC = ACC + X[I%+50] - X[I%-51]
290   Y[I%] = ACC
300 NEXT I%
310 '
320 END
```

TABLE 15-2